

Vardaan Mehandiratta

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EDUCATION

University of Waterloo

Waterloo, ON

Bachelor of Applied Science in Computer Engineering

2025 – 2030

- Scholarships: President's Scholarship of Distinction, David Johnston International Student Entrance Scholarship, International Student Entrance Scholarship

EXPERIENCE

Full Stack Developer

Mississauga, ON

APi Group (Habtech Division)

Apr 2026 – Present

- Built a field reporting platform with floor-plan mapping and technician dispatch workflows using **Next.js**, **C#**, **.NET**, **AWS**, and **DynamoDB**, streamlining inspections across **1,000+ site assets** and cutting manual coordination by **90%**.
- Architected a company-wide payroll and timesheet system with multi-stage approval workflows, Azure authentication, and batch processing, replacing spreadsheet-based operations for **500+ employees**.
- Built an AI documentation automation tool using web scraping, automated screenshot generation, and Claude API, reducing manual documentation effort by **75%** after product releases.

Software Engineering Intern

Remote, Canada

Bharat Denim

May 2025 – Aug 2025

- Built backend features for an inventory management system serving **10,000+ customers**, optimizing SQL queries and resolving data-flow issues to improve reliability across inventory and order workflows.
- Refactored MySQL and PostgreSQL schemas for inventory, orders, and product data by normalizing tables, adding indexes, and tightening constraints, contributing to a **15%** reduction in daily processing time and improved scalability.
- Worked on a backend team of 5 by driving Git workflows, including branching strategy, PR reviews, and merge conflict resolution, improving code quality and reducing integration issues across the codebase.

PROJECTS

CXR-Triage (CXC AI Hackathon) | *Next.js, TypeScript, Python, PyTorch, Supabase*

- Spearheaded a full-stack medical triage system in a **36-hour sprint**, integrating a Python inference engine with a Next.js frontend to prioritize critical cases for radiologists.
- Trained a custom **CNN** using PyTorch with data augmentation and hyperparameter optimization, achieving **80% classification accuracy** on chest X-ray datasets.
- Implemented privacy-focused data controls in PostgreSQL using row-level authorization and an append-only audit trail to protect sensitive updates.

AIInterview (McHacks 13) | *React, TypeScript, Tailwind CSS, Gemini API, ElevenLabs, Supabase*

- Built a high-fidelity interview simulation platform with a real-time dashboard tracking **11+ performance metrics**, reducing average page load time by **0.8s** and improving completion rates.
- Integrated Gemini Pro and ElevenLabs APIs for automated technical assessments, achieving an **88%** speech-to-text success rate during a 24-hour sprint.

AI-Powered Netflix Recommendation Engine | *React, TypeScript, Tailwind CSS, Supabase*

- Built an intelligent questionnaire app with AI-driven conversational flow to generate personalized recommendations based on user mood, reducing content search time by **48%**.
- Architected a context-aware feedback loop to refine results and prevent duplicates, achieving **92% recommendation accuracy** with a scalable Supabase backend.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, C#, Go, C++, C, SQL (MySQL, PostgreSQL)

Frameworks: React, Next.js, Node.js, Express.js, .NET, FastAPI, Tailwind CSS

DevOps/Tools: AWS, Azure, Docker, Git, CI/CD, MongoDB, Supabase, Vercel, DigitalOcean, Jest, Postman

Libraries: PyTorch, pandas, NumPy